COVID-19, GDP, and Higher Education: Impacts and Interactions

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***Abstract*—This project analyzes state-level economic, higher educational, and public health data in the United States across three phases of the COVID-19 pandemic: pre-COVID (2018–2019), during COVID (2020–2021), and post-COVID (2022–2024). An entity-relationship model (ERM) and relational database schemas were constructed to support structured storage and efficient querying of panel data. Data wrangling and aggregation were performed using SQL, followed by exploratory and correlation analysis in Python. Key variables include GDP, education investment, employee compensation, and COVID-19 mortality during the pandemic period. Given the limited variability and scope of the dataset, the analysis does not identify clear or robust trends across periods. The observed relationships between economic factors, educational inputs, and mortality outcomes vary substantially by state and remain statistically weak. The project demonstrates an end-to-end analytical pipeline from database design to visualization, while underscoring the impact of data limitations on empirical interpretation.**

Keywords—Covid, GDP, Education, Mortality

I. Introduction

A. Background

The COVID-19 pandemic has caused unprecedented disruptions to both the United States economy and its higher education system. While the economic consequences of the pandemic have been widely studied, the effects on higher education and the relationship between GDP and educational outcomes remain less explored. Understanding these relationships is critical for policymakers and institutions to mitigate long-term impacts.

This paper addresses two key research questions in the United States context: (1) how COVID-19's impact on GDP affects higher education metrics, and (2) how GDP levels influence higher education outcomes. To answer these questions, we construct a relational database linking GDP, higher education indicators, and COVID-19 impact measures, and employ statistical analyses to evaluate their relationships.

Although previous studies have examined the economic impact of COVID-19 and higher education separately, few have quantitatively assessed their interconnections using structured data and systematic analysis methods. By designing an efficient database schema, performing data preprocessing, and applying statistical and regression analyses, this study provides a comprehensive assessment of these interactions in the United States.

The remainder of this paper is organized as follows: Section II describes the database design, data preparation, and analytical methods; Section III presents the experimental results; and Section IV discusses the findings and provides concluding remarks.

B. Datasets Introduction

1) State-level General Statistics

a) Annual Population (U.S. Census Bureau): The annual population dataset is obtained from the U.S. Census Bureau's Population Estimates Program, which provides official population estimates for U.S. states [1]. The original spreadsheet reports population counts for multiple geographic levels, including the United States as a whole, major census regions, the District of Columbia, Puerto Rico, and all 50 states, with separate columns reporting annual population estimates for multiple years. For the purpose of preprocessing in this study, only state-level records are retained. Rows corresponding to non-state geographic areas—such as national totals, census regions, and Puerto Rico—are removed to ensure consistency with other state-level datasets used in the analysis. From the remaining records, the total annual population estimate (PopulationEst), together with the associated StateName and Year, is extracted. The original dataset is reshaped from a wide format, in which population estimates for multiple years appear in a single row, into a long-format state–year dataset.

b) Annual Gross Domestic Product (U.S. Bureau of Economic Analysis): The annual gross domestic product (GDP) dataset is obtained from the U.S. Bureau of Economic Analysis (BEA) through the Interactive Data application (iTable), which provides official economic statistics for U.S. states and other

geographic aggregations [2]. The original dataset reports annual GDP values measured in current U.S. dollars, together with related indicators such as total employment, and is disseminated in spreadsheet format over a long historical period. From the remaining state records, as with the Annual Population table, non-state records (e.g., Puerto Rico) were excluded from all datasets, and only the annual GDP value (GDP) and the associated total employment estimate (TotalEmp) are retained, together with the corresponding GeoFips, StateName, and Year identifiers. Other economic breakdowns and industry-level components available in the original BEA data are not retained, as they are not required for subsequent analysis.

2) Education-Related Data

a) Higher-Education Finance Data (SHEEO SHEF): The higher-education finance dataset is obtained from the State Higher Education Executive Officers Association (SHEEO) through the annual State Higher Education Finance (SHEF) data release, which provides state-level financial and enrollment statistics for U.S. public higher education systems [3]. The original dataset contains a wide range of fiscal-year indicators, including various cost adjustments, revenue components, state and local support measures, and full-time-equivalent (FTE) enrollment. For the purpose of preprocessing in this study, all other financial adjustment variables and revenue-related metrics provided in the original dataset are excluded, because they are not necessary for the scope of this project. Moreover, two variables are extracted from the original SHEF dataset: Total State Support, representing state government funding for public higher education, and Net FTE Enrollment, representing full-time-equivalent student enrollment. These variables, together with the corresponding state name (State) and fiscal year (FY), constitute the subset of attributes required for the relational database and subsequent analysis. Specifically, fiscal-year values (FY) are standardized as Year to maintain consistency with other datasets. The data are then converted into a structured long-format state–year table containing four attributes—StateName, Year, Support, and Enrollment.

b) Degrees Conferred Data (NCES IPEDS): The degrees conferred dataset is obtained from the Integrated Postsecondary Education Data System (IPEDS) maintained by the National Center for Education Statistics (NCES), U.S. Department of Education [4]. Specifically, the dataset corresponds to IPEDS degree completions table 318.40, which reports the number of degrees and certificates awarded annually by U.S. postsecondary institutions, categorized by institution type and degree level. The original data are provided in spreadsheet format and cover multiple academic years. For the purpose of preprocessing in this study, other institutional characteristics and detailed institutional identifiers available in the original IPEDS dataset are excluded, as they are not required for the scope of this project, and only aggregated state-level summaries by institution type and degree level are retained. The dataset is organized by four key attributes: SchType, Year, Level, and ConfNo. SchType indicates the type of institution, Year denotes the relevant calendar or fiscal year, Level represents the category of educational attainment (including Below Associate, Associate, Bachelor, Master, and Doctor), and ConfNo records the number of degrees or certificates conferred for each category. The resulting dataset follows a structured long-format design, where each record corresponds to a unique combination of institution type, year, and degree level.

c) Compensation of Employees in Educational Services (U.S. Bureau of Economic Analysis): The compensation of employees in educational services dataset is obtained from the U.S. Bureau of Economic Analysis (BEA) through the Interactive Data application (iTable), which provides state-level industry economic statistics for the United States [5]. The original dataset reports annual employee compensation for the educational services industry across multiple geographic aggregations, including national totals, multi-state regions, and individual states. For the purpose of preprocessing in this study, the data are restricted to state-level observations only. Records corresponding to geographic units above the state level are removed to ensure consistency with other state-based datasets used in the analysis. From the remaining observations, one record is retained for each state–year combination, indexed by GeoFips, StateName, and Year, with EmpComp representing total employee compensation in the educational services industry, measured in current U.S. dollars.

3) Covid-19-Related Data

a) Respiratory and COVID-19 Mortality Data (CDC): State-level mortality data are obtained from the Centers for Disease Control and Prevention (CDC) National Center for Health Statistics (NCHS) through the CDC WONDER Online Database, covering three respiratory-related disease categories: chronic lower respiratory diseases, influenza and pneumonia, and COVID-19 [6], [7], [8]. For the purpose of preprocessing in this study, mortality records are organized at the state–year–disease level. Each observation is indexed by StateAbbr, Year, and Disease, and includes two outcome measures: Deaths, representing the total number of disease-related deaths, and MortalityRate, representing the age-adjusted mortality rate per 100,000 population. Both variables are retained to support different analytical purposes. Consistent with CDC methodological guidance, age-adjusted mortality rates are preserved for descriptive analysis and cross-state comparisons within the same disease category and are not aggregated across diseases, whereas death counts may be used for temporal or period-based comparisons. To enable meaningful comparison across different pandemic periods, a period-specific disease selection strategy is adopted. Mortality from chronic lower respiratory diseases and influenza is used for years prior to 2020, while COVID-19 mortality data are used from 2020 onward. This approach avoids conflating distinct disease regimes while allowing comparison of respiratory-related

mortality patterns across pre-COVID, COVID, and post-COVID periods.

b) School Closure Data (CDC): The school closure dataset is obtained from the Centers for Disease Control and Prevention (CDC) COVID-19–Related School Closures in the United States database, which reports school-level closure and reopening events during the COVID-19 pandemic for the period 2020–2022 [9]. The original dataset is compiled from public health surveillance sources and is distributed in spreadsheet format, containing detailed information on individual school closure incidents across U.S. states. For the purpose of preprocessing in this study, other descriptive and administrative variables present in the original CDC dataset are excluded, as they are not required. For the scope of this project.each record is treated as a single school closure event and is identified by a unique event identifier (UniqID). From the original dataset, the following attributes are retained: StateAbbr, indicating the U.S. state where the school is located; SchoolName, representing the name of the school; Year, denoting the calendar year of the closure; DayClosure and DayReopened, recording the dates on which the school closed and reopened; and EpiWeek, indicating the epidemiological week associated with the closure event. The resulting dataset preserves a school-level, event-based structure and constitutes the finalized preprocessed version of the school closure data used in this study.

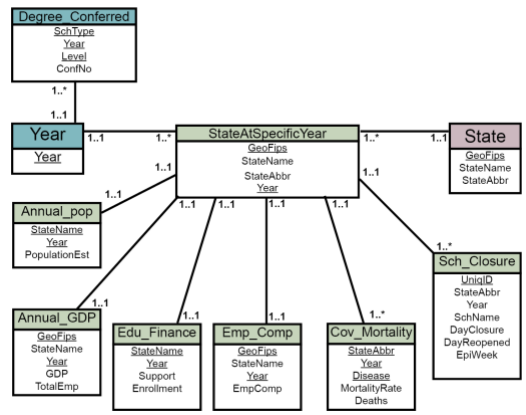
II. Methods

A. Database Design and Data Preparation Methods

1) ER Model

The Entity–Relationship (ER) model serves as the conceptual foundation of the database used in this study. As shown in Fig. 1, the model integrates multiple state-level datasets by organizing them around a central associative entity, StateAtSpecificYear, which links each state to a specific year. This design provides a unified structure that supports consistent joins across all tables when performing longitudinal or cross-state SQL analyses.

Fig. 1. ER Model



The StateAtSpecificYear entity contains the attributes GeoFips, StateName, StateAbbr, and Year, and acts as the primary hub that connects to several domain-specific entities through one-to-one (1..1) or one-to-many (1..*) relationships. Each entity captures a different aspect of state-level information:

- Annual_pop records population estimates (StateName, Year, PopulationEst), supporting demographic analysis.
- Annual_GDP stores economic indicators (GeoFips, StateName, Year, GDP, TotalEmp).
- Edu_Finance provides education-related financial and enrollment information (StateName, Year, Support, Enrollment).
- Emp_Comp captures employment composition (GeoFips, StateName, Year, EmpComp).
- Cov_Mortality stores epidemiological outcomes (StateAbbr, Year, Disease, MortalityRate, Deaths).
- Sch_Closure contains school-closure-related variables (UniqID, StateAbbr, Year, SchName, DayClosure, DayReopened, EpiWeek).

The supporting entities State, Year, and Degree_Conferred connect to the central schema to maintain referential integrity and provide additional contextual information. Their relationships (e.g., State ↔ StateAtSpecificYear as 1..1 to 1.., Year ↔ other entities as 1..1 to 1..) ensure consistent temporal and geographical referencing throughout the dataset.

Overall, this ER model enables a clean, normalized representation of heterogeneous public datasets while preserving their natural time-state structure. By consolidating all domain-specific attributes through a shared key structure (GeoFips and Year), the schema ensures efficient join operations and supports scalable query performance evaluation in later stages of the study.

2) Normalization

The database was designed in third normal form (3NF) to minimize redundancy and ensure all attributes are fully functionally dependent on their primary keys. The Sch_Closure table stores school closure data, including state abbreviation,

school name, year, dayclosure and dayreopened, and epidemiological week, with UniqID as the primary key. Annual state population estimates are stored in the Annual_Pop table with attributes StateName, Year, and PopulationEst, keyed by (StateName, Year). The Edu_Finance table captures state-level educational data, including Enrollment and Support, also keyed by (StateName, Year). The Degree_Conferred table contains the number of degrees conferred by school type and level each year, with (SchType, Year, Level) as the primary key. Disease-specific mortality data are stored in Cov_Mortality, keyed by (StateAbbr, Year, Disease). Employee compensation is recorded in Emp_Comp with full dependency on (GeoFips, Year); the partial dependency GeoFips → StateName was resolved via the State reference table (GeoFips, StateName, StateAbbr). All tables follow 3NF, enabling efficient querying and analysis.

3) Create Base Tables

Based on the normalized schema, base tables were created in MySQL with appropriate primary keys and data types to reflect the logical design.

B. Methods to Optimize SQL Query Performance

1) View

A unified state_year_panel view was created to serve as the primary analytical table at the state–year level and to integrate information from multiple heterogeneous source tables into a single analytical structure. In this view, several attributes are directly joined from base tables that are naturally defined at the state–year level using (state, year) or (GeoFips, year) as their primary keys. These include state identifiers from the State reference table (StateName and StateAbbr), economic indicators from Annual_GDP (GDP and TotalEmp), employee compensation from Emp_Comp, higher-education enrollment and state support from Edu_Finance, and population estimates from Annual_Pop. Because these tables are already organized at the state–year granularity, they can be incorporated into the panel without additional aggregation.

Variables originating from tables whose primary keys are not defined at the state–year level are incorporated through intermediate aggregation views. Specifically, respiratory mortality data from Cov_Mortality, which are recorded by state, year, and disease category, are aggregated to the state–year level and included in the panel as covid_deaths via the view v_respiratory_mortality_state_year. Similarly, school closure information from Sch_Closure, which is originally recorded at the school-event level, is summarized in the view v_closure_state_year to obtain, for each state and year, the total number of schools closed and the cumulative number of closure days. By integrating directly joinable state–year tables with selectively pre-aggregated indicators, the state_year_panel view combines demographic, economic, educational, health, and policy-related variables into a single denormalized dataset covering the period from 2018 to 2024. All variables in the panel are uniquely defined at the state–year level, which avoids redundancy and eliminates repeated aggregation in downstream analytical queries.

2) Index

To improve the performance of queries involving school closure information, an index was created on the Sch_Closure table. This table differs from other datasets in the database because it records closure events at the school level, resulting in a substantially larger number of records compared with tables organized at the state–year level. Consequently, queries that filter or join on closure-related attributes would otherwise require scanning a large volume of rows. To address this issue, a composite index was defined on the attributes (StateAbbr, Year, DayClosure, DayReopened). These columns correspond to the most common filtering, grouping, and join dimensions used when summarizing school closure events at the state–year level or retrieving closure duration information. Indexing this combination of attributes enables the database engine to more efficiently identify relevant records, thereby reducing query execution time for pre-aggregation views and subsequent analytical queries. Because other tables in the database are already structured at a much coarser state–year granularity and contain comparatively fewer records, no additional indexes were introduced for those tables within the scope of this project.

3) Constraint and Trigger

To enforce domain integrity for key geographic identifiers, declarative constraints were initially considered to restrict invalid values in the GeoFips attribute during data insertion and update operations. The intended constraint logic was designed to exclude non-state geographic codes, such as national-level or multi-state aggregate identifiers, which are incompatible with the state-level analytical framework of this study. However, under the database system environment used in this project, declarative constraints alone were not sufficient to fully enforce the required validation rules at execution time. As a result, triggers were employed as a complementary mechanism to implement the intended domain checks procedurally. These triggers validate GeoFips values during insertion and update operations and prevent records with invalid geographic identifiers from entering the base tables. This combined approach ensures that domain integrity is maintained consistently across all relevant tables while remaining compatible with the system-level capabilities of the database platform.

4) Insert and Update

All datasets in this project were initially populated through bulk loading from CSV files rather than through individual insertion statements. This approach is more efficient for initializing large tables and is consistent with typical data ingestion workflows for external datasets. Nevertheless, insert and update operations are considered in this study for two methodological purposes. First, they provide a controlled mechanism for validating the integrity-enforcement logic implemented through constraints and triggers, ensuring that newly introduced records would be subject to the same validation rules as the bulk-loaded data. Second, insert and update operations are used to verify that changes made to base

tables are correctly reflected in previously defined views, including intermediate aggregation views and the unified state_year_panel view.

C. Methods to Analyze Results

Correlation Analysis: Pearson correlation coefficients were computed to quantify the linear association between COVID-related (or respiratory) deaths and GDP across three periods: pre-COVID (2018–2019), COVID (2020–2021), and post-COVID (2022–2024). This analysis highlights changes in the strength and direction of the GDP–mortality relationship over time.

Difference-in-Differences (DID) Analysis: An illustrative DID approach was applied to compare states with high and low COVID impact. States were classified into treatment (high impact) and control (low impact) groups based on median COVID deaths. The DID framework estimates the differential effect of the pandemic by contrasting the changes in GDP, higher education enrollment, and state education support before and during COVID between the two groups. This method helps isolate the effect of the pandemic from underlying temporal trends.

III. Experimental Results

A. Optimized SQL Query Performance Results

1) View

To assess the performance benefits of view-based query optimization, school closure data were selected as a representative test case. Because the Sch_Closure table records individual school-level closure events, computing state-level statistics directly from this raw dataset requires repeatedly evaluating closure durations for each school and aggregating these values across schools and years. Executing such computations on the event-level table introduces substantial processing overhead.

To reduce this cost, a two-stage summarization structure was introduced. The first view, v_school_closure_days, consolidates multiple closure events for the same school by computing the total number of closure days for each state–school–year combination. This step eliminates repeated date-difference calculations during subsequent analysis. Building on this result, the second view, v_state_closure_summary, further aggregates these school-level summaries to the state–year level, producing two indicators essential for analysis: the total number of closure days and the number of schools that experienced at least one closure in each state and year. Since this view is already organized at the state–year level, queries that previously required scanning large numbers of event-level rows now operate on a substantially smaller and more structured dataset.

To evaluate performance, closure statistics for the 2020–2021 period were computed using two approaches:

- An event-level query, which directly processes the raw Sch_Closure table, and
- A summary-view–based query, which retrieves the same statistics using the state-year summary view v_state_closure_summary.

When executed on the raw table, the query must compute closure durations for each individual event and perform grouping operations across a large number of rows. In contrast, the summary-view approach reduces the computation to a simple aggregation over pre-summarized state–year totals. Although the absolute runtime varies slightly across repeated executions due to system-level factors, these variations are minor and do not alter the overall conclusion.

Table X compares the execution time of the event-level query and the summary-view–based query for the 2020–2021 period.

TABLE I. Comparison of Query Execution Time With and Without the Summary View (v_state_closure_summary)

Query Scenario	Description	Execution Time (Seconds)
Raw Query (No View)	*Computes closure durations directly from event-level Sch_Closure records*	*5.265*
Summary-View Query	*Retrieves pre-summarized state–year totals from v_state_closure_summary*	*0.234*

As expected, the version using v_state_closure_summary executed significantly faster, demonstrating the effectiveness of hierarchical summarization in reducing query complexity and improving performance. Table X reports the execution times for both methods and shows that the summary-view approach achieves an approximate 95.6% reduction in execution time relative to the event-level query.

This experiment illustrates that restructuring complex event-level data into layered summary views can dramatically improve analytic query performance, especially when repeated computations and large granular datasets are involved.

2) Index

To assess indexing performance, the original event-level school-closure query (5.265 seconds without indexing) was used again as the baseline. To mitigate the computational overhead associated with this baseline query, a composite index was created on the attributes most frequently used in filtering and grouping operations—state abbreviation, year, and the closure and reopening dates. This allows the query engine to locate relevant records more efficiently, avoiding full table scans.

Table X compares the execution time of the event-level school-closure query before and after applying an index for the 2020–2021 period.

TABLE II. COMPARISON OF QUERY EXECUTION TIME WITH AND WITHOUT INDEXING

Query Scenario	Description	Execution Time (Seconds)
Raw Query (No Index)	*Direct computation from Sch_Closure (event-level data)*	*5.265*
Indexed Query	*Same event-level computation after applying a composite index*	*2.516*

After applying the index, the same event-level query was executed again. The indexed version completed in 2.516 seconds, representing an approximate 52.2% reduction in execution time compared with the unindexed execution. Although the performance gain is smaller than achieved using the summary-view approach (which eliminates event-level processing entirely), indexing still provides a meaningful improvement by accelerating row lookups and reducing join-processing cost.

This experiment demonstrates that, in datasets with large numbers of granular records, carefully selected composite indexes can substantially improve query performance—particularly in scenarios where repeated filtering or joining on the same attributes is required.

3) Trigger

To evaluate the effectiveness of the trigger-based validation mechanism, the Annual_GDP table was used as a test case. The trigger trg_annual_gdp_geofips_check was designed to prevent inserting or updating records that contain invalid GeoFips values, specifically "00000" or region-level codes that follow the pattern "9%000". These geographic codes represent national or multi-state aggregates and are not valid state identifiers for the purposes of the database.

Two test queries were executed to confirm the trigger's behavior. First, an attempt was made to insert a new record containing GeoFips = '00000'. Second, an update operation was applied to change an existing valid state code to GeoFips = '90000'.

As shown in Table X, in both cases the trigger correctly intercepted the operation and returned an error message indicating that the GeoFips value is invalid.

TABLE III. TRIGGER VALIDATION RESULTS FOR DETECTING INVALID GEOFIPS INPUTS

Query Scenario	Description	Error Message
Invalid Insert	*INSERT INTO Annual_GDP VALUES ('00000', 2024, 12345.6, 5000000)*	*Error Code: 1644. Invalid GeoFips: non-state geographic code*
Invalid Update	*UPDATE Annual_GDP SET GeoFips = '90000' WHERE GeoFips = '01000'*	*Error Code: 1644. Invalid GeoFips: non-state geographic code*

These results demonstrate that the trigger successfully enforces data integrity by blocking prohibited geographic codes before they enter the Annual_GDP table. This validation occurs automatically at the database level, ensuring consistent data quality regardless of how records are inserted or updated.

4) Insert and Update

To evaluate the behavior of INSERT and UPDATE operations within the database and to verify that changes made to a base table are automatically reflected in dependent views, the Annual_Pop table was used as a demonstration case. A temporary view, v_demo_2025_population, was defined to display population values for New York for the years 2024 and 2025, enabling direct observation of how modifications propagate through the relational structure.

Before inserting any new record, the view showed only the existing 2024 population estimate for New York (Table X).

TABLE IV. POPULATION VALUE FOR NEW YORK IN 2024 BEFORE INSERTING THE 2025 RECORD.

StateName	Year	PopulationEst
New York	*2024*	*19867248*

A new record for 2025 was then inserted by copying the 2024 value. Immediately afterward, an UPDATE operation increased the inserted 2025 population by 20 percent using the expression: SET PopulationEst = PopulationEst * 1.2.

This multiplicative adjustment provides a clear numerical indicator of whether the update was successfully applied. When the view was queried again, it displayed both the original 2024 value and the newly inserted 2025 value, with the latter correctly reflecting the 20% increase (Table Y).

TABLE V. UPDATED POPULATION VALUES FOR NEW YORK IN 2024–2025 AFTER INSERTING AND MODIFYING THE 2025 RECORD.

StateName	Year	PopulationEst
New York	*2024*	*19867248*
New York	*2025*	*23840698*

Because the view definition performs no additional computation beyond selecting from the base table, the updated record appeared automatically without modifying the view itself. This behavior confirms that views in MySQL dynamically

reflect current base-table contents, making them reliable tools for downstream analysis even when base tables undergo insertions or updates.

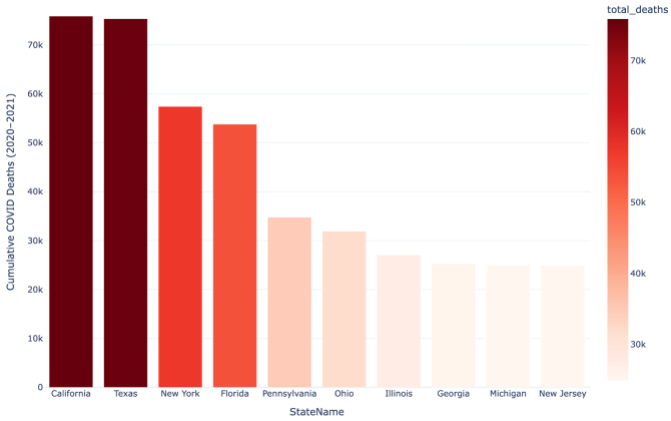
- Overall, the experiment demonstrates two key outcomes: both the insertion of a new state–year record and the subsequent update were successfully executed and validated through observable numerical changes; and
- the view v_demo_2025_population correctly and immediately incorporated these modifications, illustrating MySQL's consistent view maintenance over underlying relational data.

B. *Graphical performance results*

TABLE VI. GDP, ENROLLMENT, AND COVID DEATHS TREND (ALABAMA AS AN EXAMPLE)

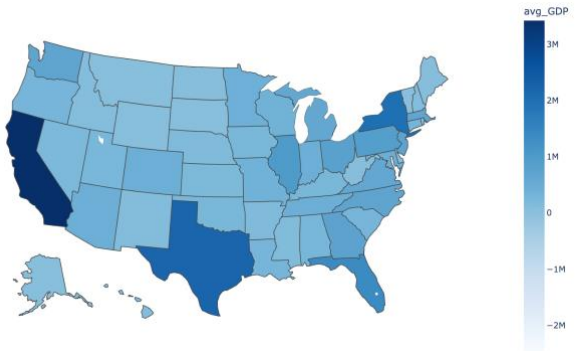
The temporal trends of GDP, higher education enrollment, and COVID-19-related deaths across 2018–2024. GDP in the United States demonstrates steady growth during 2018–2019, followed by an accelerated increase beginning in 2020. Enrollment exhibits a similar overall trend, with a slight decline at the onset of the pandemic, followed by gradual recovery. COVID-19-related deaths increased sharply during the early phase of the pandemic and subsequently declined, reflecting both the severity and eventual control of the outbreak.

TABLE VII. TOP 10 STATES BY COVID DEATHS DURING THE PANDEMIC



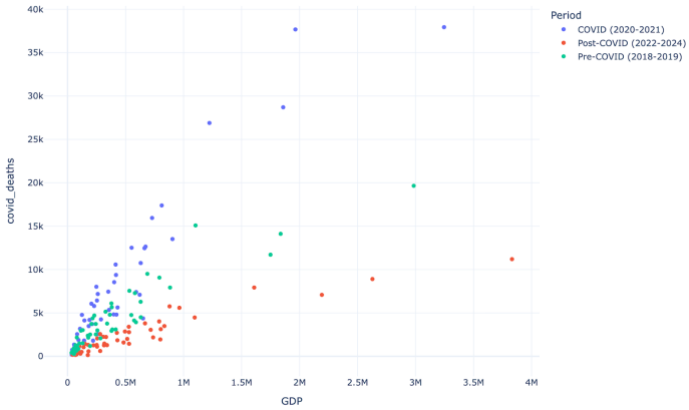
The top ten states by cumulative COVID-19 deaths during 2020–2021. California, Texas, New York, and Florida rank highest, indicating that these states experienced the most severe mortality burden in absolute terms. This emphasizes the geographic concentration of pandemic impact and its overlap with major economic centers.

TABLE VIII. AVERAGE GDP BY STATE



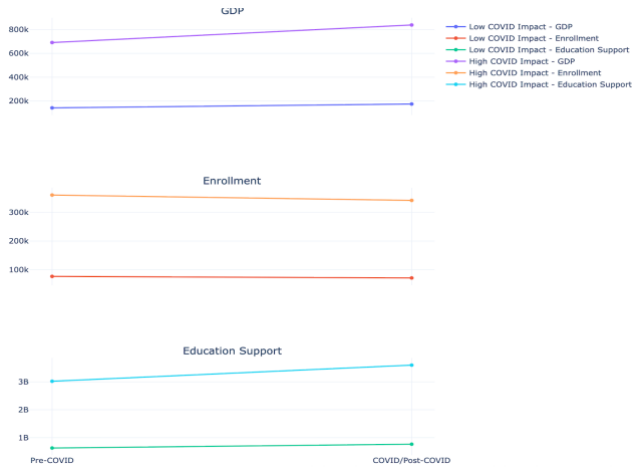
The average GDP by state, highlighting California, Texas, Florida, and New York as the largest economies. High-GDP states are primarily concentrated along the coasts and in Texas, which provides context for interpreting subsequent pandemic outcomes in economically significant regions.

TABLE IX. CORRELATION: GDP VS COVID DEATHS BY STATE AND PERIOD



The scatter plot of GDP versus COVID-related deaths across different periods. During the COVID period (2020–2021), states with GDP above 0.25 million exhibited markedly higher mortality compared to the pre-COVID period (2018–2019), whereas states with GDP below 0.25 million showed mortality levels like the pre-COVID period. In the post-COVID period (2022–2024), mortality declined substantially, falling below pre-COVID levels across all GDP ranges, indicating a reversal of the elevated mortality observed during the pandemic in higher GDP states.

TABLE X. ILLUSTRATIVE DID-STYLE MEAN COMPARISON



The illustrative DID-style mean comparison of GDP, Enrollment, and Education Support between states with high and low COVID impact. It can be observed that the average GDP in high COVID impact states increased noticeably during the COVID period (2020–2021), whereas low impact states exhibited relatively stable GDP. Similarly, respiratory/COVID deaths increased significantly in the high impact states compared to the low impact states. Enrollment showed a slight decline in high impact states, while remaining largely unchanged in low impact states. These results highlight the differential effects of the pandemic across states with varying COVID severity.

IV. DISCUSSION/CONCLUSION

The COVID-19 pandemic exerted heterogeneous effects across U.S. states. High-GDP, high-impact states experienced elevated mortality alongside accelerated GDP growth, with minor declines in higher education enrollment, suggesting that economic performance remained resilient despite temporary educational disruptions. The results indicate a positive association between GDP and enrollment resilience, while extreme health shocks can produce short-term educational setbacks in severely affected regions.

Limitations of this analysis include the use of chronic respiratory and influenza mortality as pre-pandemic proxies versus COVID-19 mortality post-2020, which may introduce measurement inconsistencies. Additionally, the assessment of educational impacts is limited to state funding and school closure data, potentially omitting other influential factors. Future research should incorporate broader datasets to enhance robustness and better isolate the effects of the pandemic on both economic and educational outcomes.

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